

**a Shared evaluation embedding**

3,964 cells across eight held-out species



● Arabidopsis thaliana  
 ● Fragaria vesca  
 ● Brassica rapa  
 ● Catharanthus roseus  
 ● Gossypium bickii  
 ● Gossypium hirsutum  
 ● Triticum aestivum  
 ● Eutrema salsugineum

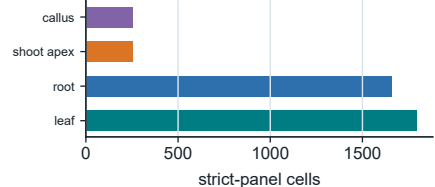
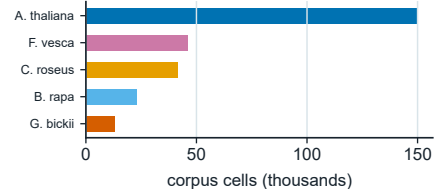
**b Ontology states in the same coordinates**

State structure is separated from species identity

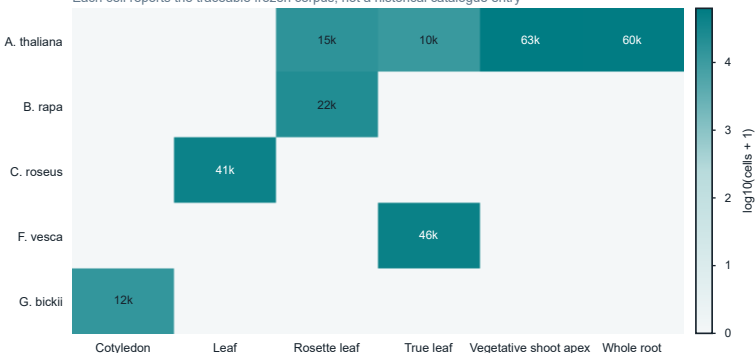
● other  
 ● mesophyll  
 ● meristem or stem cell niche  
 ● epidermis  
 ● cell cycle state  
 ● vascular stele  
 ● xylem  
 ● cortex  
 ● root cap  
 ● phloem  
 ● endodermis

**c Frozen corpus profile**

272/132 cells, 209,406 genes

**d Training data are distributed across species and organs**

Each cell reports the traceable frozen corpus, not a historical catalogue entry

**e A protocol-aware annotation contract**

The model records assumptions that change the meaning of an annotation result



The frozen profile, inner-fold choice, target support and deployment head are never presented as one metric.